The two finite chiral atlases: recovered results and source corrections

Human Compression project

This manuscript gives a source-wide disposition of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382. The first atlas and all printed entries of the second atlas have complete accepted finite evidence. Several printed numbers and one automorphism-group label require correction. The larger domain stated in Section 4 also requires an explicit boundary: the printed roster has 79 groups and omits one group satisfying that section's literal condition. We do not invent a census for that omitted group.

The appended complete component proofs and the exact source ledger specify what is acquired. Standard regular and chiral reconstruction theorems retain the source's attribution to McMullen–Schulte and Schulte–Weiss. Exact finite GAP group operations, the named group constructions and automorphism classifications, and completeness of the earlier regular-cover atlas remain stated inputs. No present website is asserted to have identical historical bytes. This synthesis repeats no group census.

Definitions and equivalence. We use the source's ranked-poset definition, including the diamond axiom and strong flag connectivity. Its Sections 2–3 group criteria are retained with their complete subset intersection property. The accepted prefix-coordinate manuscript supplies the interval/difference subgroup dictionary, reflection criterion and complete stabilizer enumeration; it does not replace polytope recognition by adjacent product orders alone. A colour-preserving isomorphism of flag graphs is a polytope isomorphism. Reversing all colours gives duality, which is not additionally identified in the atlas counts below.

In the first atlas, the two enantiomorphic markings are identified as one unrooted chiral polytope. In the second atlas, a generating rotation marking is taken modulo the full automorphism group of its abstract group; its mirror is counted separately. The reflection criterion implies that this mirror involution has no fixed chiral class. Thus division by two there forgets orientation, not duality. Regular columns count full regular automorphism groups, not rotation groups extracted from the chiral search.

Theorem 1 (Theorem 3.1 and the source algorithm)**.** *Let the actual flag graph of a quotient polytope be the right-coset graph of $N \leq W = \langle r_0, \ldots, r_{d-1} \rangle$, where the r_i are its specified flag adjacencies. Its colour automorphism group is $N_W(N)/N$. The quotient is chiral precisely when $[W : N_W(N)] = 2$ and all r_i lie outside $N_W(N)$. There is at most one index-two subgroup H avoiding all r_i. For that H, the possible chiral stabilizers are precisely $N \lhd H$ not normal in W, subject to the actual polytope condition. Conjugate stabilizers N and $r_0 N r_0$ describe the same unrooted quotient.*

Proof. A colour automorphism is determined by the image Na of N and must send Nw to Naw. This is well-defined exactly when a normalizes N. The resulting action is free, with $[W : N_W(N)]$ orbits. For an index-two normalizer, adjacency by r_i exchanges the two orbits exactly when r_i is outside it. The subgroup avoiding every generator is the kernel of the unique possible parity homomorphism $r_i \mapsto 1 \in C_2$. A subgroup normal in that H but not in W has normalizer exactly H, so it has the two stated conjugates. The complete coset and geometric interface proofs are in *Chiral quotients from one parity subgroup.* □

The source's core-free filter is optional: the kernel of the flag action is $\mathrm{Core}_W(N) = N \cap r_0 N r_0$. It is not legitimate to discard nontrivial cores when excluding arbitrary chiral quotients. Nor do we infer that every monodromy quotient is a regular polytope. Minimal faithful-cover language is used only with its specified flag-action interface.

Theorem 2 (The first atlas and Table 1)**.** *Relative to the acquired complete regular-cover atlas of orders at most 2000, excluding 1024 and 1536, there are exactly 48 rank-three and eight rank-four chiral quotients, with distinct duals counted separately. The four Table 1 markings and their reversals are the eight rank-four examples. Their displayed cover and rotation-group structures are correct. The vertex figures of the last two rows have types* $\{6, 18\}$ *and* $\{6, 9\}$ *and are among the verified rank-three chiral examples.*

Proof. The accepted full source parser supplies 9,212 nondegenerate marked covers. The acquired product argument excludes a bond of order two; ranks one and two cannot be chiral. All normal stabilizers allowed by the preceding criterion are considered, including nontrivial cores. Three exact completed intervals partition the source roster. Their 8,986 normal candidates leave 756 after the necessary base-face intersection test.

Literal coset transport verifies 704,608 edges. Of the 756 candidates, 360 have full colour isomorphisms to the 56 accepted examples, covering all of them; the other 396 have explicit negative witnesses, namely 350 diamond failures and 46 disconnected flag residues. The examples themselves have complete face-poset, chain, diamond and residue checks, and complete rooted colour comparisons proving distinctness and chirality. This is the accepted *Identifying the quotient flag graphs* component, with no reliance on an interrupted candidate or on abstract database identifiers for exclusions.

Table 1's six structural names were proved on the actual cover groups and rotation quotients by literal generator-edge transports, normal subgroups, complements and commuting factors. In particular the last two rows really have the same named cover and rotation groups. The source convention is $|D_9| = 18$; its parentheses are preserved. The complete names and certificates are in *The six group descriptions in the small chiral atlas*.

For the remaining vertex-figure sentence, keep the base vertex in either accepted rank-four flag graph and take its component using colours $1, 2, 3$. Renumbering those colours gives the vertex-figure flag graph by the polytope section definition. The saved components have respectively 216 and 108 flags. Complete bijections to accepted rank-three entries 44 and 39 preserve all 648 and 324 coloured edges. Thus the sections are the asserted chiral polytopes. Every vertex has flags from both global flag orbits (an adjacency of colour one preserves the vertex and exchanges those orbits), so the chiral automorphism group is vertex-transitive. The base-vertex calculation therefore applies to all vertex figures of each example. □

Theorem 3 (All printed almost-simple entries)**.** *For the 79 distinct named group models in Tables 2–7, occurring in 88 rows, the regular and oriented chiral counts are as follows. All rows agree rank by rank except the regular rank-three entry for* $\mathrm{PSL}_2(27)$*, which is 26 rather than 27, and the chiral rank-three entry for* $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$*, which is 46 rather than 26.*

rank	3	4	5	6	7	8	*total*
regular	8319	810	61	16	3	1	9210
oriented chiral	15056	1890	28	0	0	0	16974

The oriented chiral classes form 8487 mirror pairs.

Proof. The accepted regular application acquires 14 earlier complete censuses and 65 complete exact executions. Its full automorphism actions are verified common actions on the named models; it does not use an inner action in place of the full automorphism group. The acquired records bind their actual drivers, models and outputs, with complete terminal rank lists; the

older records' runtime-file-trace limits remain explicit. A separate fixed-endpoint calculation gives $312/12 = 26$ for the exceptional regular row.

The accepted chiral application acquires 72 complete whole searches, the complete 25-branch S_9 union, and six complete fixed-first-class unions. A fixed first list is proved to contain exactly one representative of each relevant conjugacy class. Unions require its literal equality, not just matching class indices. Each centralizer recursion is complete by the acquired stabilizer theorem. Heredity and strict subgroup growth supply its all-rank stopping boundary. Full automorphisms act freely on generating markings, so division of the complete inner-orbit count by $|\mathrm{Out}(G)|$ is valid separately at each rank and type. An independent literal positive certificate already distinguishes 46 chiral classes for the exceptional unitary row, without using its census completeness or the outer-order division.

The accepted applications join these results to every printed occurrence through literal named models. Nine repeated occurrences are removed only after their group identifications; they contribute 95 regular and 50 chiral counts to the raw family sums. Their removal gives the displayed totals. No source count was a filter or premise of these enumerations. □

Family totals and the observations in Section 6. The exact family sums, retaining repeated occurrences between families, are

Table	2	3	4	5	6	7
regular	1536	804	3903	1846	1202	14
oriented chiral	4780	1710	1882	3998	4242	412

Besides the two row corrections, the printed aggregate arithmetic in Table 3 (1702), Table 4 (2050), and Table 5 (1840 regular) is corrected by these sums. Section 6's global 9205, 17114 and 8557 become respectively 9210, 16974 and 8487. Its PSL_2-family rank-at-least-four counts are 45 regular and 338 chiral, replacing 51 and 506. The comparative comments about sporadic, unitary and Suzuki groups remain statements about this finite roster and its stated convention. They do not compare all groups or all genera. As in the source's aggregate, $\mathrm{Aut}(J_2)$ is excluded: its order 1209600 lies beyond both stated cutoffs. The older 5265 regular objects identified duals; the historical Conder genus counts, MAGMA timings and website descriptions remain attributed context, not new calculations or interchangeable counting conventions.

Orders and named automorphism groups. The complete source application identifies every printed group and automorphism label through acquired constructor/classification and literal transport or outer-preimage routes. For $S \le G \le \mathrm{Aut}(S)$ with S simple, restriction identifies $\mathrm{Aut}(G) \cong N_{\mathrm{Aut}(S)}(G)$: the unique simple socle is characteristic, its centralizer is trivial, and a normalizing automorphism extends by conjugation. The actual preimages and small outer normalizers, not order equality alone, supply the names. The accepted $J_2 : 2$ join additionally has an actual outside involution complement.

The source's order 443510 for M_{22} is corrected to 443520, and its order 7800 for $\mathrm{PSL}_2(25).2$ to 15600. The automorphism label $P\Gamma L_3(3)$ for $\mathrm{PSU}_3(3)$ is corrected to $P\Gamma U_3(3)$, of order 12096. In contrast $P\Gamma L_3(3) = PGL_3(3)$ has order $(26 \cdot 24 \cdot 18)/2 = 5616$.

The historical-domain statement requires a correction. The abstract uses a 900000 cutoff and speaks of the group Γ appearing in the 1985 ATLAS. Section 4 instead gives a one-million cutoff and explicitly requires only the simple socle S to appear. Under that latter socle condition, the accepted complete simple/outer roster and the explicitly retained CTblLib transcription of the 1985 contents give 80 almost-simple isomorphism classes. Exactly 79 have order below 900000; these are precisely the printed models. The additional group is $\mathrm{Sp}_4(4)$, of order 979200, whose simple socle has ordinary ATLAS page 44. Thus the printed roster does

not exhaust the larger Section 4 domain. This corrects the coverage claim, rather than silently replacing its domain by the printed list.

The historical ATLAS transcription is an explicit premise, not a claimed direct inspection of a book scan. We do not infer that every outer extension had its own printed ATLAS table, nor what an unavailable historical program executed. No polytope count, zero result, or new census for $\mathrm{Sp}_4(4)$ is asserted. The recovered theorem is the complete printed 79-group atlas with the corrected entries, plus this exact larger-domain omission.

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